

Erdős Problem #644

EP-644

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $f(k, r)$ be minimal such that if $A_1, A_2, \dots$ is a family of sets, all of size k , such that for every collection of r of the A_i s there is some pair $\{x, y\}$ which intersects all of the A_j , then there is some set of size $f(k, r)$ which intersects all of the sets A_i . Is it true that $f(k, 7) = (1 + o(1))\frac{3}{4}k$? Is it true that for any $r \geq 3$ there exists some constant c_r such that $f(k, r) = (1 + o(1))c_r k$?

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